Wooah Choi

Machine Learning Engineer, Search, Ranking and Recommendation

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wooahchoi.com | backenddeveloper.tistory.com | London, UK | No visa sponsorship required until 2029

Machine learning and search engineer with nearly three years delivering production NLP, search, ranking and recommendation systems, now finishing an MSc in Artificial Intelligence at Queen Mary University of London. Built BERT-based query understanding, large-scale indexing and ranking systems across legal, insurance, library and government domains, taking Supreme Court litigation search from 5+ minutes to 8 seconds and improving Millie's Library ranking quality by 38%. Built GridFlow Trade, a UK day-ahead price forecasting product selected for QMUL QIncubator, on 65,000+ settlement periods, including the point-in-time feature pipeline, the MLOps around it and a linear-programming decision layer.

Skills

Machine Learning	Python, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, feature engineering, model selection, classification, regression, optimisation, time-series forecasting, backtesting
Search, Ranking & NLP	BERT, DistilBERT, ColBERTv2, Hugging Face Transformers, semantic and vector search, FAISS, BM25, RRF, ranking and recommendation, named entity recognition, query understanding, LLMs, RAG
Evaluation & Infrastructure	offline evaluation, benchmark construction, A/B testing, statistical testing, ablation studies, experiment tracking, model versioning, SQL, PostgreSQL, Oracle, Java, C++, TypeScript, Docker, MLflow, Jenkins, GitLab CI, AWS (EC2, S3, RDS), Linux, Spring, FastAPI, REST APIs

Work Experience

GridFlow Trade, London | *Co-Founder, AI and Backend* | gridflowtrade.com Nov 2025 – Apr 2026

Co-founded a UK energy trading startup. Selected for the QMUL QIncubator programme and pitched to an investor panel, April 2026.

- Owned the full lifecycle: ingestion, validation, training, versioned registry, batch and API serving, monitoring, rollback and retraining. **Found that target leakage and train/serve skew had inflated the reported accuracy** (MAE £7.4 in training against £21.9 when served) and rebuilt a single point-in-time feature pipeline shared by training and inference, fixed at the D-1 09:30 decision time. Later caught the same pattern twice more in my own monitoring and dashboard, where forecasts cached after the day began were being scored, **which could have stopped the rollback rule firing on a real degradation**
- XGBoost MAE £24.2/MWh against £34.2 naive on a 90-day held-out window (**29% lower**, mean of 5 seeds), and **23% below naive over a 12-month walk-forward** with monthly refits, better in 11 of 12 months. Benchmarking Naive, Ridge, LightGBM, XGBoost and a bidirectional LSTM on one leak-free split in MLflow showed the tree models and Ridge tied within seed noise, so the gains came from feature and evaluation design rather than model choice; ablation put NESO day-ahead forecasts at **22% of the reduction**
- Backtested the battery strategy (linear-programming dispatch, SciPy HiGHS) at realised prices over 360 days: **63% of perfect-foresight revenue against 46% for a naive forecast**, 61% vs 42% after trading and degradation costs, loss-making days 14% to 5%, maximum drawdown £5.3k to £2.0k. Found an optimiser flaw that sold the starting charge for free every day, £3.5k/day of phantom revenue, before it reached the results
- Retraining runs unattended on a weekly schedule, or early when the served model falls behind naive, gated on seven checks against the production model and the baseline and registered to MLflow with the artefact SHA-256; deployment installs only on a hash match and **a mismatch was refused in production**. **Automatic rollback running in production** reverts to the last verified artefact when daily realised accuracy falls behind naive. Two versions served side by side behind a FastAPI selection API that hot-reloads without a restart and holds the previous model if a new one fails to load
- Ingested Elexon and NESO feeds into PostgreSQL on hourly lock-protected jobs under Alembic schemas, with hourly data-quality checks covering **46 and 50-period clock-change days**, PSI drift and per-route latency. Audited 3.7 years of stored data against the Elexon API and found a collector date bug that had corrupted **1,634 price and 1,634 system-margin records** on every British Summer Time day. CI on every push: 80+ unit tests, 43 Playwright end-to-end tests in four timezones, **each regression test confirmed to fail on the old code first**

Konan Technology Inc, Seoul | *Software Engineer, AI Solution Development Team* Nov 2020 – Jun 2023

Production search, ranking and NLP for more than four enterprise clients across banking, insurance, the judiciary and consumer tech, plus fourteen shorter projects. Owned the full lifecycle from requirements and architecture through model integration, CI/CD, deployment and handover.

- **Supreme Court of Korea** | Next-generation electronic litigation search and analysis [Java, Spring, Jenkins, GitLab, Python, Linux, Docker, C++, PyTorch, BERT, Oracle]
  - Built the query understanding layer: fine-tuned BERT-based NER over case numbers, statutes and party names, a **4%+ gain in entity extraction**, with speech act analysis and semantic role labelling for intent. Entities typed at index time so queries hit the right field, not the whole corpus
  - Rebuilt the Java backend and indexing pipeline: search latency **5+ minutes to 8 seconds**, 375+ misanalysed cases resolved, facet search 9.8s to 0.06s
  - Semantic and vector search for query expansion and typo correction, related and recommended keywords via LDA on search logs, Jenkins CI/CD on GitLab, and search analytics dashboards for administrators
- **Millie's Library** | Book search, ranking and user pattern analysis [C++, Python, MySQL, Linux]
  - Automated indexing for 100M+ records, with a volume structure designed to scale beyond 1B
  - Built the ranking layer from nothing, combining field match weights with 7 and 30-day engagement signals. **Ranking quality improved 38%** against a fixed test query set
  - Personalised recommended search terms with no labelled data, grouping accounts on search time and estimating term demand from history

- **Korea Eximbank** | Records management search with high availability [Java, Oracle, Linux]
  - Dual-server search architecture for a state policy bank: single-writer indexing with forced failover, index replication on recovery and L4 load balancing, so search stayed available without the nodes diverging
  - Sized index partitioning against a 4 million record, terabyte-scale corpus, and delivered disclosure-level access control over a four-level classification scheme
  - Took on project management and client negotiation. The specified redundancy topology could not fail over, so I built the case for a non-technical stakeholder and the design was changed
- **Heungkuk Fire & Marine Insurance** | Main website integrated search renewal [Java, JavaScript, Oracle, Linux]
  - Rebuilt the integrated search for a major Korean insurer, full stack from indexing to the customer-facing search page, with auto-completion and recommended search terms driven by customer query logs

Projects

Multi-Domain Neural Search Engine and RAG Grounding Evaluation Feb 2026 – Sep 2026
MSc Information Retrieval coursework, QMUL · Team project; grounding and statistics added independently

- Built a 6-domain search engine over MS MARCO, SciFact, FiQA, TREC-COVID, NFCorpus and SCIDOCS: one dense index per domain on **BAAI/bge-base-en-v1.5** (msmarco-bert for general), a fine-tuned DistilBERT query router, hybrid BM25 + FAISS-HNSW with RRF fusion and ColBERTv2 late-interaction reranking
- Added a RAG grounding framework: 6 causal context-perturbation conditions measuring whether an answer depends on the retrieved passages or on what the model already knew
- Built the **A/B testing infrastructure** behind it: two-proportion z-tests, paired Welch's t-tests, Benjamini-Hochberg FDR correction and Cohen's h/d effect sizes across 21 metrics, validated with a 5-step ablation ladder

Multimodal VQA and LLM Fine-Tuning 2026
MSc coursework, QMUL · Neural Networks and NLP; Conversational Agents and Dialogue Systems

- Built a LLaVA-style multimodal model from scratch: frozen LongCLIP-ViT-B/32 vision encoder, Qwen3-0.6B language model and a trainable projector, with LoRA added in a second phase. Caption restoration 99.83%, base VQA 61.36%, **77.85% final** after caption-aware late fusion, seed ensembling and option-shuffle test-time augmentation
- Compared full supervised fine-tuning against LoRA on Gemma-3-270M: **LoRA reached F1 33.00% against 32.16% for full SFT while training 2.75% of the parameters**. Also fine-tuned Qwen3.5-0.8B with LoRA, raising task accuracy from 10% to 20%

MSc Dissertation | QMUL · Supervisor: Paulo Rauber Feb 2026 – Sep 2026
Causal Cue Retrieval and Greedy Route Execution: A Mechanistic Analysis of a Transformer Policy in a Fixed MiniGrid T-Maze

- Trained and evaluated causal Transformer, parameter-matched LSTM and MLP policies with PPO in a partially observable MiniGrid task, using cue-stratified metrics to distinguish genuine memory use from shortcut policies hidden by aggregate success
- Developed mechanistic interpretability audits using cue ablation, retrieval blocking, opposite-cue memory patching and norm-preserving source swaps, showing that correlational attribution could misidentify the internal source causally controlling a decision
- Diagnosed failures across seeds, checkpoints and action-selection rules: sampling recovered 4/7 apparent argmax failures to $S_{\min} = 0.970$–1.000, and the clean mechanism signature held in only **3 of 6 checkpoints** that solved the task equally well, so equal behaviour did not imply equal mechanism across retrains

Research Agent Lab | github.com/hncpyj/research-agent-lab 2026 – Present
Protocol Before Code: Stress-Testing Control Boundaries in Autonomous Research Agents. Choi, W. Technical report, 2026.

- Built an autonomous research agent, then designed *Protocol Before Code* after watching it silently substitute a different experiment for the one that had been approved. The architecture freezes approved intent in a machine-checkable protocol, binds trusted components by content hash and checks scientific conformance separately from software correctness
- In fault-injection and holdout evaluation **0/18** drifted protocols passed and **18/18** behaviour-changing component mutations were blocked before execution; with the integrity check disabled **15/18** of the same mutations escaped the remaining stack, isolating that safeguard's marginal value

Education

MSc in Artificial Intelligence | Queen Mary University of London, London, UK Sep 2025 – Sep 2026

- **Expecting Distinction**. Core modules: Information Retrieval, NLP, Neural Networks and NLP, Machine Learning, Reinforcement Learning, Conversational Agents and Dialogue Systems, *Ethics, Regulation and Law in Digital Decision Making*

BSc in Computer Science | Korea National Open University, Seoul, South Korea Mar 2021 – Aug 2023
Double Major: Statistics and Data Science

- Cumulative GPA 3.3/4.5 (87%), UK 2:1 equivalent. Core modules: Multivariate Analysis, Regression Models, Bayesian Data Analysis, Data Mining, Forecasting Methods, Discrete Mathematics (A+), Linear Algebra (A+), Deep Learning (A)

BEng Environmental Engineering | Catholic University of Korea, Seoul Mar 2013 – Feb 2018

Publications and Academic Service

Oct 2026 *Relevant Is Not Enough: Expert-Audited Decision-Grade Web Evidence for Financial Fact Verification.* Park, H., Kim, E., **Choi, W.**, & Zubiaga, A. FinNLP 2026 @ EMNLP 2026, Budapest. **Accepted (poster), to appear.** Contribution: annotated 500 of the 750 claim–passage pairs under a blinded four-class protocol, including the 250-pair double-annotated block behind the paper's Cohen's kappa.

Aug 2026 Reviewer, FinNLP 2026 @ EMNLP 2026 *(ACL SIG-FinTech)*